

Results outline corintreespotters

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Research questions

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species?
4. How did growth change across years, and how did it relate to climate?

Results

1. Summary basic information
 - (a) Coringtreespotters includes eleven with leaf phenology data spanning nine years (2016 to 2024) for a total of 326 leafout and 289 colored leaves observations, yielding 278 complete growing seasons (GS) for 49 individuals.
 - (b) 2017 was the coolest year (21.53°C), and 2020 was the warmest (23.07°C)
 - (c) 2020 and 2022 had moderate droughts that ramped up to severe droughts in September. 2016 had a moderate drought starting in June, ramped up to severe in July, then to extreme in August and September. There were no droughts in the other years.
 - (d) GSL ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021)
 - (e) GS thermal sum ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD, (*T. americana* in 2016)
 - (f) The earliest leafout day of year was 08-Apr (*A. flava* in 2021) and the latest, 15-May (*C. ovata* in 2020). The earliest colored leaves day of year was 15-Jul (*A. flava* in 2021—same individual with earliest leafout) and the latest, 06-Nov (*T. americana* in 2016)
2. Does tree growth increase with longer growing seasons?
 - (a) No clear trend of growing season length effect across our eleven species and season predictors.
 - (b) Longer thermal season increased growth for no species and decreased growth for *B. alleghaniensis* ($\beta = -0.23$, UI_{90} [-0.36, -0.1]) (Figure 1a and e) (Table S1).
 - (c) No species responded to longer calendar season (Figure 1b and f) (Table S1).
3. Did a later start and end of season increase growth?
 - (a) An earlier start of season increased growth for no species and decreased growth for *T. americana* ($\beta = 0.15$, UI_{90} [0.03, 0.27]) (Figure 1c and g) (Table S1).
 - (b) A later end of season increased growth for no species and decreased growth for *B. alleghaniensis* ($\beta = -0.09$, UI_{90} [-0.16, -0.01]) (Figure 1d and h) (Table S1).
 - (c) The estimates were unchanged when using the full or restricted datasets (SOS $n = 326$ and EOS $n = 289$ vs $n = 278$ Figure S4b), or when using budburst instead of leafout as the start of the growing season (Figure S5c).
4. How did tree growth change across years and tree ids?
 - (a) We did not find any clear effect of year on growth (Figure 2) and there is no clear differences of growth looking across species (Figure S1 and Figure ??)
 - (b) Growth did not change across any provenances (Figure ??) (Table ??)

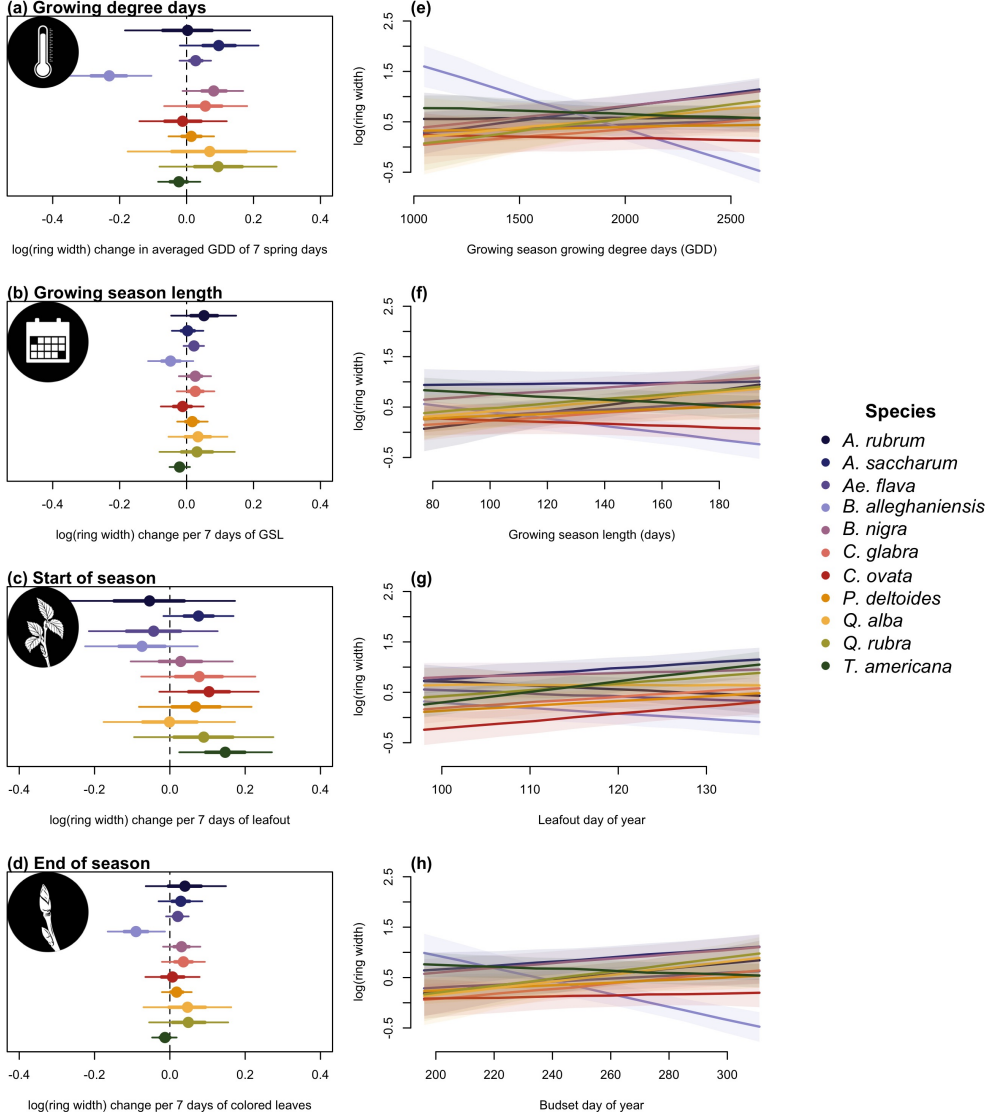


Figure 1: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and coloredLeaves and the models estimates shows ring width change per 7 average spring days GDD (176.87) (a). GSL is the number of days between leafout and coloredLeaves (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through coloredLeaves (d). All predictors were fit using the same model and number of observations, with only the independent variable changing. The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

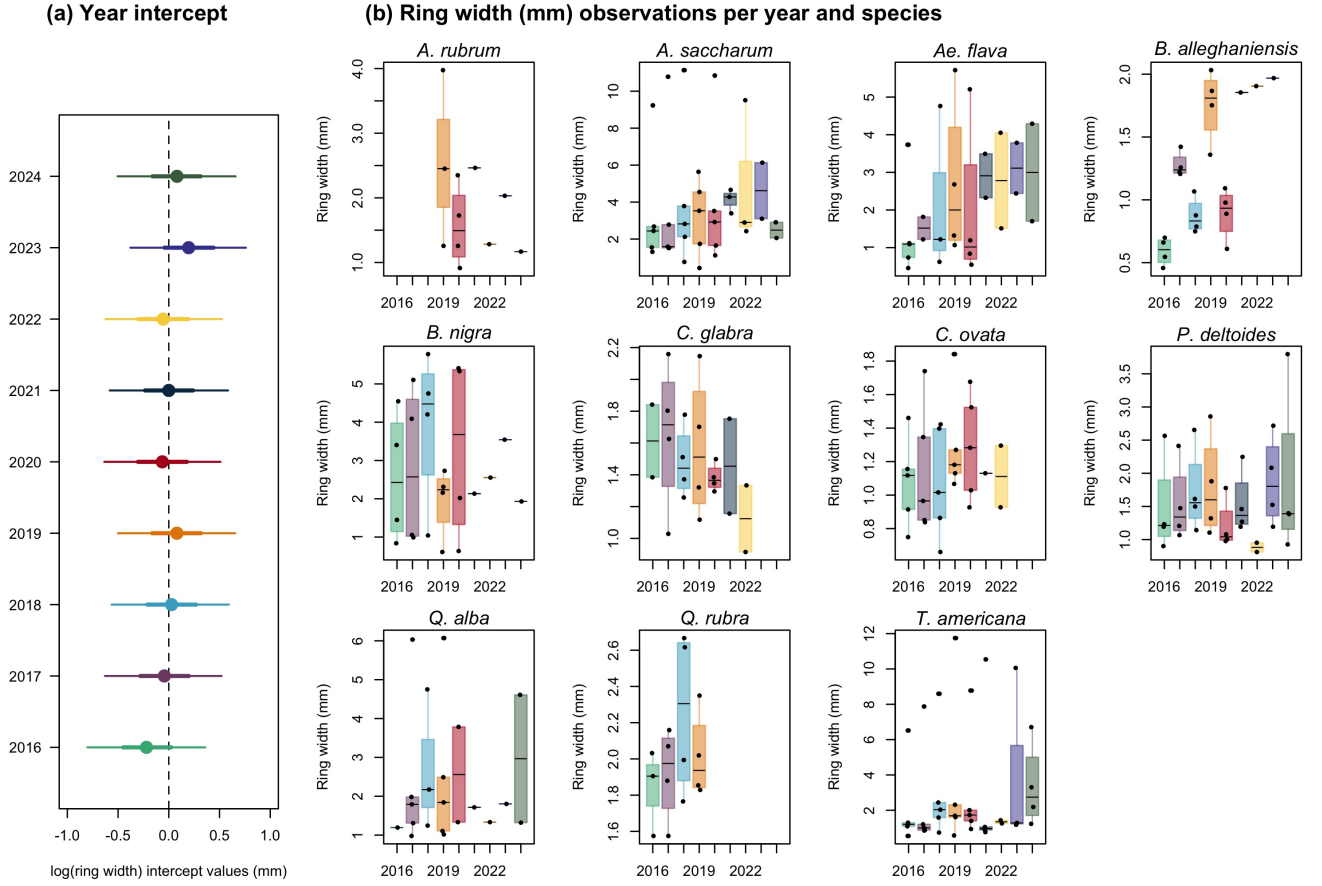


Figure 2: We show tree ring widths (mm) across years as model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI).

Supplemental results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Coringtreepotters. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
<i>A. rubrum</i>	0.00 [-0.18, 0.19]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.17]	0.04 [-0.06, 0.15]
<i>A. saccharum</i>	0.10 [-0.02, 0.21]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.09]
<i>Ae. flava</i>	0.03 [-0.02, 0.07]	0.02 [-0.01, 0.05]	-0.04 [-0.21, 0.13]	0.02 [-0.01, 0.05]
<i>B. alleghaniensis</i>	-0.23 [-0.36, -0.10]	-0.05 [-0.12, 0.02]	-0.07 [-0.23, 0.07]	-0.09 [-0.17, -0.01]
<i>B. nigra</i>	0.08 [-0.01, 0.17]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.17]	0.03 [-0.02, 0.08]
<i>C. glabra</i>	0.06 [-0.07, 0.18]	0.03 [-0.03, 0.08]	0.08 [-0.08, 0.23]	0.04 [-0.02, 0.09]
<i>C. ovata</i>	-0.01 [-0.14, 0.12]	-0.01 [-0.08, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.07, 0.08]
<i>P. deltoides</i>	0.01 [-0.05, 0.08]	0.02 [-0.03, 0.06]	0.07 [-0.08, 0.22]	0.02 [-0.02, 0.06]
<i>Q. alba</i>	0.07 [-0.18, 0.33]	0.03 [-0.05, 0.12]	-0.00 [-0.18, 0.17]	0.05 [-0.07, 0.16]
<i>Q. rubra</i>	0.09 [-0.08, 0.27]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
<i>T. americana</i>	-0.02 [-0.09, 0.04]	-0.02 [-0.05, 0.01]	0.15 [0.03, 0.27]	-0.01 [-0.05, 0.02]

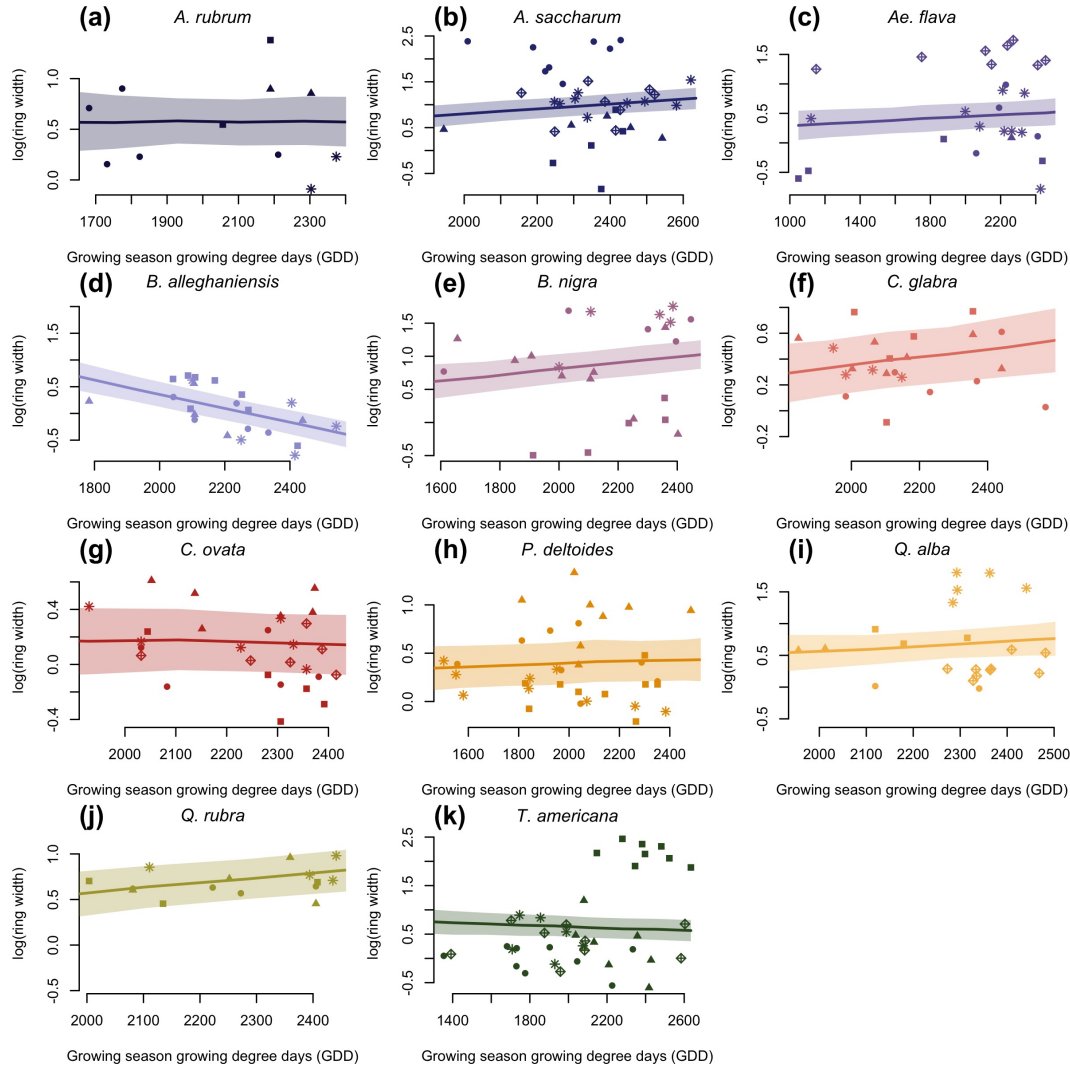


Figure S1: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

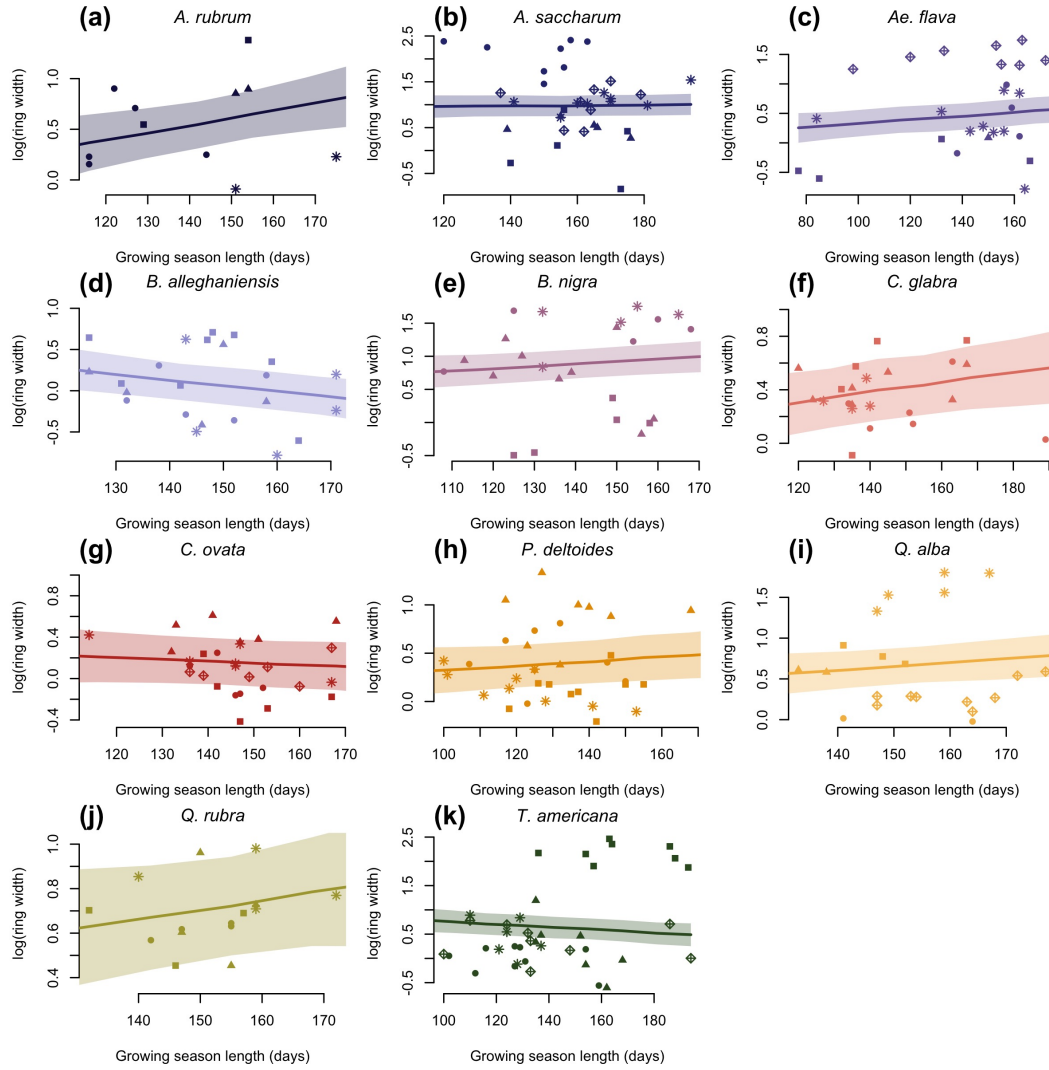


Figure S2: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

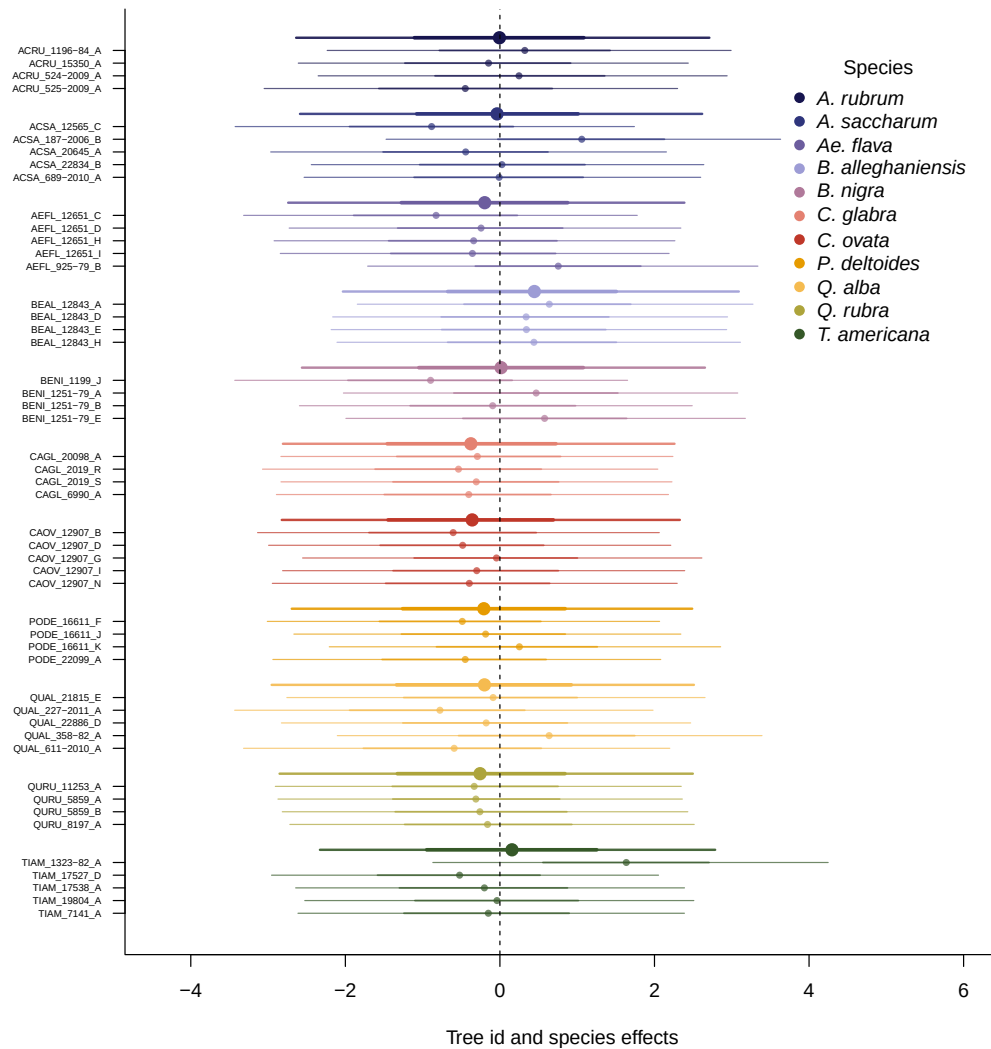


Figure S3: Intercept estimates for each group of the GDD model. The different tree id values represent the sum of the tree id, provenance, species and averaged year intercepts. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

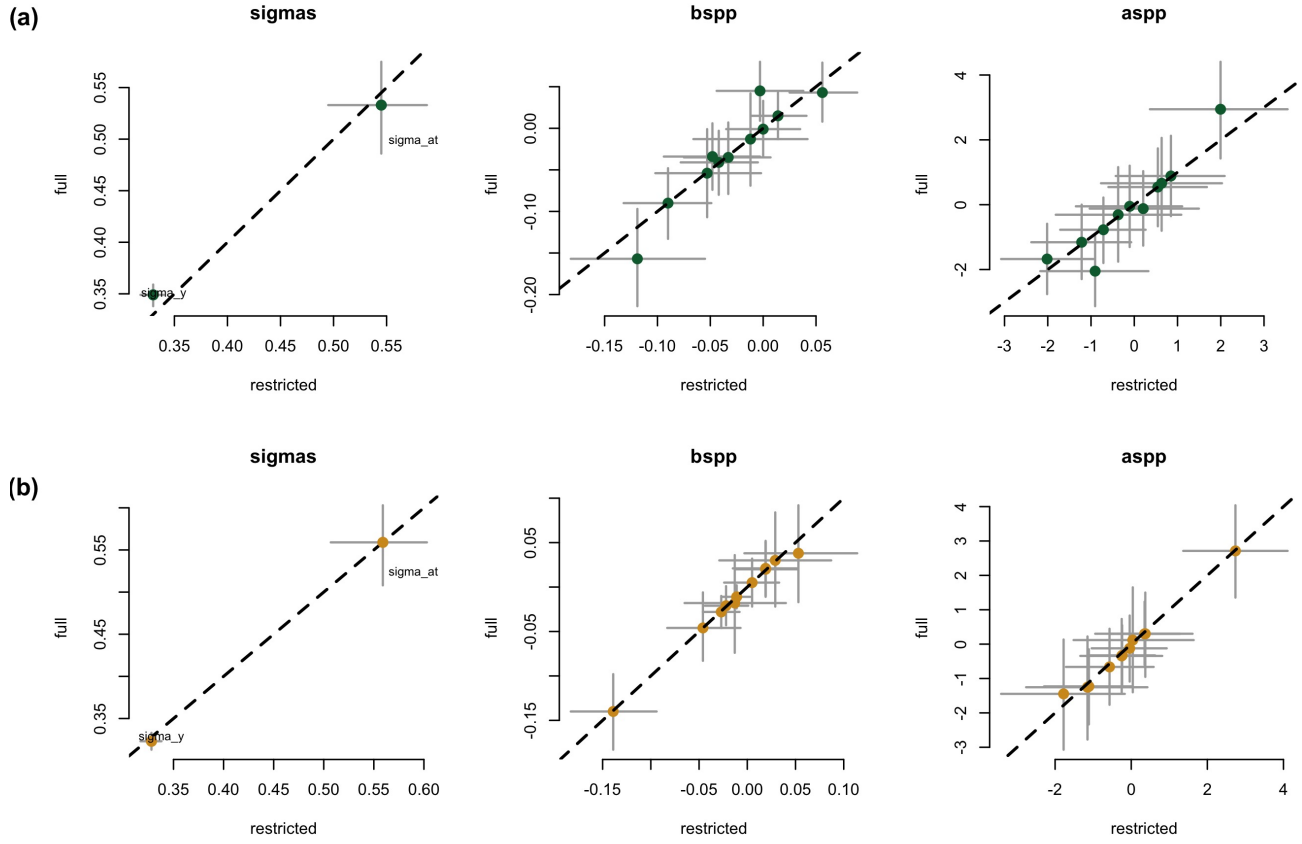


Figure S4: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for CoringTreespotter. Each panel represent the different parameters used to fit the models.

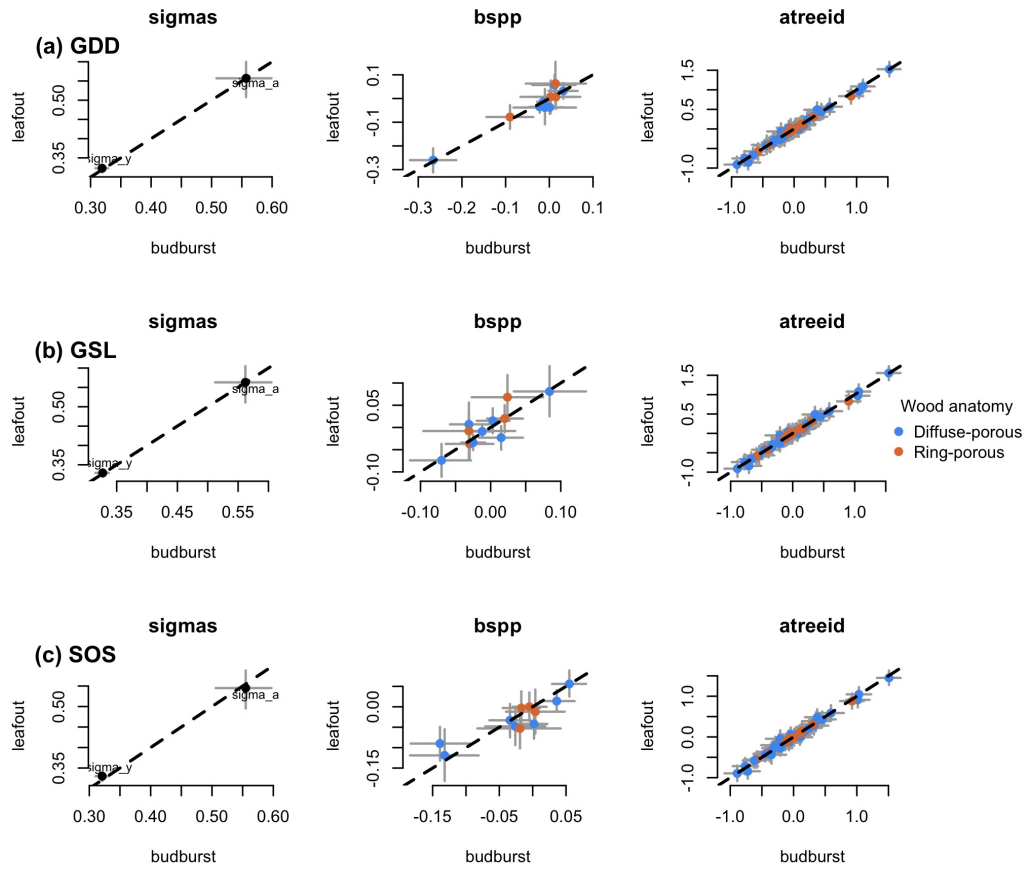


Figure S5: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS.